End-to-End Data Management Pipeline for a Recommendation System

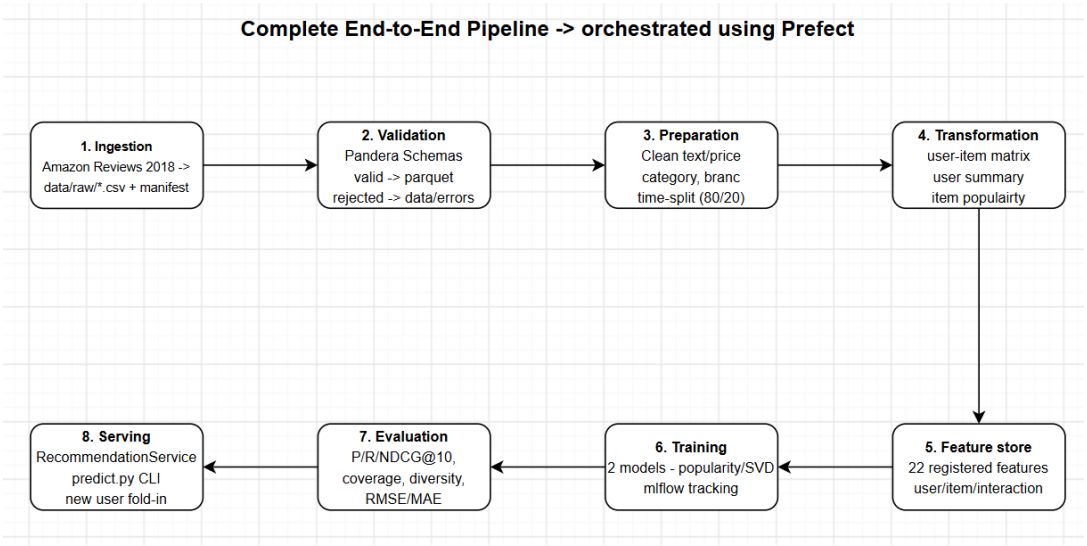
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Course: Data Management for Machine Learning
GitHub repository: https://github.com/govindsatwani/recommendation-pipeline

1. Objective and Dataset:

- The objective of this project was to build a modular, reproducible data management pipeline for a recommendation system.
- The pipeline covers data ingestion, validation, preparation, transformation, feature management, model training, evaluation, and orchestration.
- The project uses the public **Amazon Reviews 2018 - Software (5-core)** dataset and its associated product metadata from the McAuley Lab.
- The reviews provide user-item interactions and ratings, while the metadata provides item attributes.
- The pipeline derives three tables: users, items, and interactions.

2. Architecture and Tools:

- The workflow is: ***Ingestion → Validation → Preparation → Transformation → Feature Management → Training → Evaluation → Recommendation Generation***
- The pipeline is implemented in Python and orchestrated using **Prefect**.
- It uses Pandas for data processing, Pandera for validation, Parquet and CSV for data storage, scikit-learn for truncated SVD, and MLflow for experiment tracking.
- Separate modules are used for each stage to make the workflow easier to inspect and maintain.



Pipeline Architecture

3. Data Ingestion and Validation:

- The ingestion stage reads the review and metadata files and creates the user, item, and interaction tables.

- It also generates an ingestion manifest with row counts, columns, and SHA-256 hashes for provenance.
- The validation stage checks missing values, duplicates, data types, rating ranges, timestamps, and referential integrity between interactions and the user and item tables.
- Rejected records are saved in error datasets with reasons, and the results are recorded in a validation report.
- A separate fault-injection demonstration applies the checks to deliberately corrupted input.

Dataset	Input rows	Valid rows	Rejected rows
Users	1,826	1,826	0
Items	26,790	21,639	5,151
Interactions	12,805	11,984	821

Table 1. Validation results

4. Preparation, Transformation, and Features:

- Preparation cleans and standardizes text and item attributes, parses and imputes prices, and handles missing values. Interactions are split chronologically before repeated user-item pairs are deduplicated separately within each partition. This preserves an earlier training interaction even when the same user interacts with the item again in the test period.
- For the reported run, the cutoff was **2015-10-04 09:36:00 UTC**, with **9,528 training rows** and **2,359 test rows** after within-partition deduplication.
- Transformation produces user activity summaries, an item popularity table, a sparse user-item interaction matrix, and user and item index mappings.
- The custom feature layer defines **22 features**: eight user features, nine item features, and five interaction features. Examples include user average rating and activity count, item popularity and average rating, recency, preferred category, and category match. The feature catalog documents their definitions and sources.
- The current models use only a subset of these features. Interaction features are materialized for potential downstream use but are not directly used in the current ranking models.
- Non-missing helpful-vote counts must be nonnegative integers; missing counts are filled with zero during preparation.

5. Model Training and Evaluation:

- Two recommenders were trained:
    - **Popularity baseline:** ranks items by their number of training interactions.
    - **Personalized SVD recommender:** uses truncated matrix factorization over the user-item interaction matrix to generate personalized rankings.

- MLflow records model parameters, metrics, and artifacts. Evaluation uses **K = 10**, with a test rating of at least 4 treated as relevant. Previously seen training items are excluded from recommendation lists.
- The ranking evaluation covers **308 users** with training history and at least one relevant future item that is in the training catalog and was not previously seen by that user.
- RMSE and MAE are calculated on **703 test interactions** whose users and items are known from training. These rating-evaluation pairs can include repeated user-item interactions.

Model	Precision@10	Recall@10	NDCG@10	Coverage	Diversity	RMSE	MAE
Popularity	0.0052	0.0203	0.0115	0.0208	0.8536	1.3723	1.1166
SVD	0.0156	0.1040	0.0686	0.3781	0.8047	1.1972	0.9643

Table 2. Models' evaluation results

- On this offline evaluation subset, SVD performs better on precision, recall, NDCG, coverage, RMSE, and MAE. The popularity baseline has higher intra-list diversity.

6. Orchestration and Reproducibility:

- Prefect runs the pipeline stages in dependency order and records their execution states. The repository includes setup instructions, pinned dependencies, configuration, reports, and a workflow that can be run once or served on a schedule.
- After creating the Python environment and installing dependencies as described in the README, the complete pipeline can be run with: ***.venv/bin/python dags/recsys_flow.py***

7. Limitations:

The following limitations were seen in the current execution of the code:

- Evaluation is offline and excludes test interactions involving users or items absent from training. Cold-start performance is therefore not fully represented by the reported metrics.
- Ranking evaluation excludes items already seen by a user in training, while rating evaluation can include repeated user-item pairs.
- Interaction-derived aggregates use training interactions, but the historical availability of the item metadata used by the feature layer has not been established.
- The interaction data is sparse, no online A/B test was performed, and extensive hyperparameter tuning was not performed.

8. Future Improvements:

Possible future improvements include:

- Separate cold-start evaluation.
- Integrating a hybrid model that incorporates item attributes into ranking.

Execution Screenshots:

1. Setup:

```
(venv) satwani@HPE-5CG2500XP6:~/dev$ cd /home/satwani/dev/P/DMML_assignment_2/recommendation-pipeline
python3 -m venv .venv
.venv/bin/pip install -r requirements.txt
export MLFLOW_DISABLE_AGENT_HINT=1
Requirement already satisfied: uncalled-for>=0.4.0 in ./.venv/lib/python3.14/site-packages (from pydocket>=0.23.1->prefect==3.9.1->-r requirements.txt (line 11
)) (0.4.2)
Requirement already satisfied: beartype>=0.20.0 in ./.venv/lib/python3.14/site-packages (from py-key-value-aio>=0.3.0->py-key-value-aio[memory,redis]>=0.3.0->p
ydocket>=0.23.1->prefect==3.9.1->-r requirements.txt (line 11)) (0.23.1)
Requirement already satisfied: shellingham>=1.3.0 in ./.venv/lib/python3.14/site-packages (from typer>=0.15.1->pydocket>=0.23.1->prefect==3.9.1->-r requirement
s.txt (line 11)) (1.5.4)
Requirement already satisfied: mypy-extensions>=0.3.0 in ./.venv/lib/python3.14/site-packages (from typing_inspect>=0.6.0->pandera==0.35.0->-r requirements.txt
 (line 6)) (1.2.0)
Requirement already satisfied: oauthlib>=3.0.0 in ./.venv/lib/python3.14/site-packages (from requests-oauthlib->apprise<2.1.0,>=1.13.0->prefect==3.9.1->-r requ
irements.txt (line 11)) (4.0.1)
```

Environmental setup

```
(venv) satwani@HPE-5CG2500XP6:~/dev/P/DMML_assignment_2/recommendation-pipeline$ .venv/bin/python --version; cat requirements.txt
Python 3.14.4
pandas==3.1.0
numpy==2.5.4
scipy==1.18.1
scikit-learn==1.9.1
joblib==1.6.0
pandera==0.35.0
pyarrow==26.0.1
PyYAML==6.0.3
requests==2.34.2
mlflow==3.18.0
prefect==3.9.1
matplotlib==3.11.2
```

Python Dependencies

2. Data Ingestion:

```
ication_code$ .venv/bin/python -m src.ingestion.ingest
2026-11-09 20:16:14,103 | INFO | ingestion | Downloading https://mcauleylab.ucsd.edu/public_datasets/data/amazon_v2/categoryFilesSmall/Software_5.json.gz
2026-11-09 20:16:26,283 | INFO | ingestion | Downloading https://mcauleylab.ucsd.edu/public_datasets/data/amazon_v2/metaFiles2/meta_Software.json.gz
2026-11-09 20:16:43,108 | INFO | ingestion | Wrote users.csv        rows=1826 cols=2
2026-11-09 20:16:43,625 | INFO | ingestion | Wrote items.csv        rows=26790 cols=7
2026-11-09 20:16:43,655 | INFO | ingestion | Wrote interactions.csv rows=12805 cols=7
2026-11-09 20:16:43,677 | INFO | ingestion | Ingestion complete
```

Ingestion Phase


```
(venv) satwani@HPE-5CG2500XP6:~/dev/P/DMML_assignment_2/recommendation-pipeline$ cat data/raw/ingestion_manifest.json
{
  "dataset": "Amazon Reviews 2018 - Software (5-core)",
  "files": {
    "users.csv": {
      "rows": 1826,
      "columns": [
        "user_id",
        "user_name"
      ],
      "sha256": "1f146312744f31e527f05dae37aac3c282d45c4f4fa78ef14c921f39e39a07a2"
    },
    "items.csv": {
      "rows": 26790,
      "columns": [
        "item_id",
        "title",
        "brand",
        "main_category",
        "categories",
        "price",
        "description"
      ],
      "sha256": "62f617829e75989b2e893fb262e8658bb333f83f4c4a30f373c8c5bc8f5fbc1f"
    },
    "interactions.csv": {
      "rows": 12805,
      "columns": [
        "user_id",
        "item_id",
        "rating",
        "timestamp",
        "verified",
        "vote",
        "summary"
      ],
      "sha256": "346adbb13acd5e18fd8b9924af73f1863af563a2553a0b6aa720ba26fe90fbe8"
    },
    "Software_5.json.gz": {
      "sha256": "1495184a2d3a166516b1fbc5be57c5dbcc9fd44eb6be3df72a67aeb5546cd9ed",
      "bytes": 5335448
    },
    "meta_Software.json.gz": {
      "sha256": "5855d757635170c7e456da56efecb9aea2210ca399ab8cbe0d50d98f98a954d0",
      "bytes": 15236627
    }
  }
```


Ingestion Manifest

3. Data Validation:

```
e$ .venv/bin/python -m src.validation.validate
2026-11-09 20:17:19,912 | INFO | validation | Loaded users        rows=1826
2026-11-09 20:17:19,912 | INFO | validation | Loaded items        rows=26790
2026-11-09 20:17:19,912 | INFO | validation | Loaded interactions rows=12805
2026-11-09 20:17:20,496 | INFO | validation | users        valid=1826 rejected=0 {}
2026-11-09 20:17:20,609 | INFO | validation | items        valid=21639 rejected=5151 {"duplicate_row": 5151}
2026-11-09 20:17:20,613 | INFO | validation | interactions valid=11984 rejected=821 {"duplicate_row": 821}
2026-11-09 20:17:20,613 | WARNING | validation | warning interactions.repeated_user_item_pairs = 163
2026-11-09 20:17:20,613 | WARNING | validation | warning items.missing_title = 6
2026-11-09 20:17:20,613 | WARNING | validation | warning items.missing_brand = 659
2026-11-09 20:17:20,613 | WARNING | validation | warning items.missing_categories = 2063
2026-11-09 20:17:20,613 | WARNING | validation | warning items.missing_price = 20182
2026-11-09 20:17:20,613 | WARNING | validation | warning items.unparseable_price = 1290
2026-11-09 20:17:20,614 | WARNING | validation | warning items.missing_description = 3597
2026-11-09 20:17:20,614 | WARNING | validation | warning users.missing_user_name = 1
2026-11-09 20:17:20,795 | INFO | validation | Validation report written to reports/validation_report.md
```

Validation Phase

```
(venv) satwani@HPE-5CG2500XP6:~/dev/P/DMML_assignment_2/recommendation-pipeline$ head -3 data/errors/interactions_errors.csv
user_id,item_id,rating,timestamp,verified,vote,summary,error_reason
A1522TN5FVJL0Y,B00005AFI2,2.0,1032912000,False,,Why?,duplicate_row
A3C1JX8E543YBQ,B000VJTL1Y,5.0,1357948800,True,,PAINT SHOP PRO,duplicate_row
```

Validation Errors

4. Data Preparation and Transformation:

```
ication_code$ .venv/bin/python -m src.preparation.prepare; cat data/processed/split_info.json
2026-11-09 20:17:38,404 | INFO | preparation | Duplicates resolved within time partitions: 97 repeated user-item rows removed
2026-11-09 20:17:38,517 | INFO | preparation | Split at 2015-10-04: train=9528 test=2359 (cold users in test: 589 rows)
{
  "rows_in": 11984,
  "rows_out": 11984,
  "repeated_user_item_pairs_removed": 97,
  "cutoff_timestamp": 1443951360,
  "cutoff_date": "2015-10-04",
  "train_rows": 9528,
  "test_rows": 2359,
  "test_fraction_actual": 0.1985,
  "train_users": 1719,
  "train_items": 722,
  "test_rows_cold_user": 589,
  "test_rows_cold_item": 1212,
  "train_period": [
    "2000-06-26",
    "2015-10-04"
  ],
  "test_period": [
    "2015-10-05",
    "2018-08-26"
  ]
```

Preparation Phase

```
de$.venv/bin/python -m src.transformation.transform; ls -lh data/processed
2026-11-09 20:17:50,613 | INFO | transformation | user_activity_summary=1719 rows | item_popularity=722 rows | matrix (1719, 722) density=0.0077
total 8.2M
-rw-r--r-- 1 satwani satwani 412K Nov  9 20:17 interactions_clean.parquet
-rw-r--r-- 1 satwani satwani  12K Nov  9 20:17 item_index.parquet
-rw-r--r-- 1 satwani satwani  18K Nov  9 20:17 item_popularity.parquet
-rw-r--r-- 1 satwani satwani 7.1M Nov  9 20:17 items_clean.parquet
-rw-r--r-- 1 satwani satwani  452 Nov  9 20:17 split_info.json
-rw-r--r-- 1 satwani satwani  85K Nov  9 20:17 test.parquet
-rw-r--r-- 1 satwani satwani 341K Nov  9 20:17 train.parquet
-rw-r--r-- 1 satwani satwani  65K Nov  9 20:17 user_activity_summary.parquet
-rw-r--r-- 1 satwani satwani  37K Nov  9 20:17 user_index.parquet
-rw-r--r-- 1 satwani satwani  23K Nov  9 20:17 user_item_matrix.npz
-rw-r--r-- 1 satwani satwani  53K Nov  9 20:17 users_clean.parquet
drwxr-xr-x 2 satwani satwani 4.0K Nov  9 20:17 validated
```

Transformation Phase

5. Feature Management:

```
e$ .venv/bin/python -m src.features.feature_store
2026-11-09 20:17:57,544 | INFO | features | Materialized 22 features: users=1719 items=21639 interactions=9528
```

Feature Management Phase

```
ion_code$ ls -lh data/features
total 604K
-rw-r--r-- 1 satwani satwani  164 Nov  9 20:30 feature_metadata.json
-rw-r--r-- 1 satwani satwani 165K Nov  9 20:30 interaction_features.parquet
-rw-r--r-- 1 satwani satwani 367K Nov  9 20:30 item_features.parquet
-rw-r--r-- 1 satwani satwani  63K Nov  9 20:30 user_features.parquet
```

Item features

Feature	Type	Description	Formula / logic	Source
`item_popularity`	int	Number of training interactions the item received.	`count(interactions)`	train.parquet
`item_unique_users`	int	Number of distinct users who interacted with the item.	`nunique(user_id)`	train.parquet
`item_avg_rating`	float	Average rating of the item (global mean if never rated).	`mean(rating)`	train.parquet
`item_popularity_rank`	int	Rank of the item by popularity (1 = most popular).	`rank(item_popularity, descending)`	train.parquet
`item_days_since_last_interaction`	float	Recency of the last interaction with the item.	`(as_of - max(timestamp)) / 86400`	train.parquet
`item_price`	float	Cleaned price (missing prices imputed with category median).	`parse(price) -> median impute`	items_clean
`item_category`	string	Top-level product category.	`first level of the category path after 'Software'`	items_clean
`item_subcategory`	string	Second-level product category (may be empty).	`second level of the category path`	items_clean
`item_brand`	string	Brand of the item ('Unknown' if missing).	`cleaned brand text`	items_clean

Item feature definitions

User features

Feature	Type	Description	Formula / logic	Source
`user_avg_rating`	float	Average rating the user has given.	`mean(rating) over the user's training interactions`	train.parquet
`user_rating_std`	float	Spread of the user's ratings (0 for a single rating).	`std(rating)`	train.parquet
`user_activity_count`	int	Number of interactions (reviews) by the user.	`count(interactions)`	train.parquet
`user_verified_ratio`	float	Share of the user's reviews that are verified purchases.	`mean(verified)`	train.parquet
`user_days_since_last_interaction`	float	Recency of the user's last interaction.	`(as_of - max(timestamp)) / 86400`	train.parquet
`user_preferred_category`	string	Category the user interacted with most (ties: higher total rating).	`argmax count(category)`	train + items_clean
`user_preferred_brand`	string	Brand the user interacted with most (ties: higher total rating).	`argmax count(brand)`	train + items_clean
`user_avg_price`	float	Average price of the items the user interacted with.	`mean(item.price)`	train + items_clean

User feature definitions

Interaction features

Feature	Type	Description	Formula / logic	Source
`interaction_category_match`	nullable Int64	1 if category matches, 0 if it does not, missing if either value is unavailable.	`item_category == user_preferred_category`	user + item features
`interaction_brand_match`	nullable Int64	1 if brand matches, 0 if it does not, missing if either value is unavailable.	`item_brand == user_preferred_brand`	user + item features
`interaction_price_ratio`	float	Item price relative to what the user usually pays.	`item_price / user_avg_price`	user + item features
`interaction_rating_minus_user_avg`	float	How much the rating deviates from the user's average (training only).	`rating - user_avg_rating`	train.parquet
`interaction_days_before_as_of`	float	Age of the interaction at the reference date.	`(as_of - timestamp) / 86400`	train.parquet

User-item interaction feature definitions

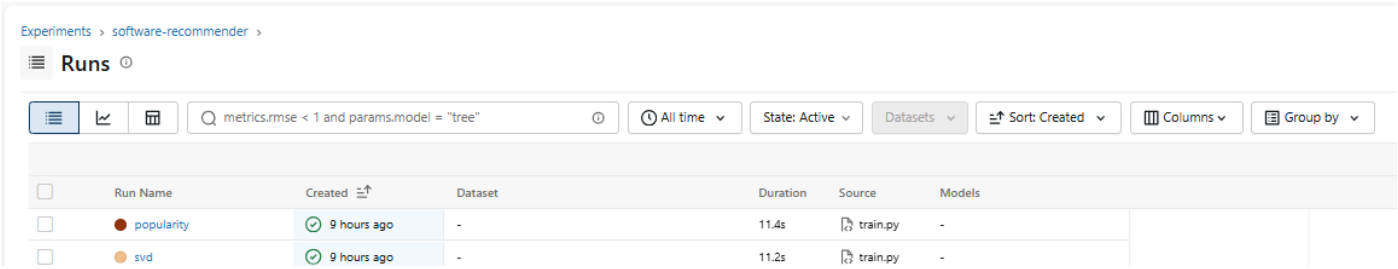
6. Training, Experiment Tracking, Evaluation:

```
e$ .venv/bin/python -m src.training.train
2026/11/09 20:18:05 INFO mlflow.store.db.utils: Creating initial MLflow database tables...
2026/11/09 20:18:05 INFO mlflow.store.db.utils: Updating database tables
2026-11-09 20:18:09,721 | INFO | training | Trained popularity in 0.07s -> popularity.joblib (run d25e2693)
2026-11-09 20:18:10,624 | INFO | training | Trained svd        in 0.70s -> svd.joblib (run af5d9a1b)
```

Training of recommendation models

```
backend-store-uri sqlite:///mlflow.db --port 5000
Registry store URI not provided. Using backend store URI.
[MLflow] Security middleware enabled with default settings (localhost-only). To allow connections from other hosts, use --host 0.0.0.0 and configure --allowed-hosts and --cors-allowed-origins.
2026/11/09 20:18:32 INFO mlflow.server.jobs.executor_registry: Available backends after discovery: docker, local
2026/11/09 20:18:32 INFO mlflow.server.jobs.executor_registry: Job executor backend 'local' passed requirements check
2026/11/09 20:18:32 INFO:     Uvicorn running on http://127.0.0.1:5000 (Press CTRL+C to quit)
2026/11/09 20:18:32 INFO:     Started parent process [50025]
/home/satwani/dev/P/DMML_assignment_2/test_by_download/recommendation-pipeline-revising_deduplication_code/recommendation-pipeline-revising_deduplication_code/.venv/lib/python3.14/site-packages/mlflow/server/fastapi_app.py:24: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
/home/satwani/dev/P/DMML_assignment_2/test_by_download/recommendation-pipeline-revising_deduplication_code/recommendation-pipeline-revising_deduplication_code/.venv/lib/python3.14/site-packages/mlflow/server/fastapi_app.py:24: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
/home/satwani/dev/P/DMML_assignment_2/test_by_download/recommendation-pipeline-revising_deduplication_code/recommendation-pipeline-revising_deduplication_code/.venv/lib/python3.14/site-packages/mlflow/server/fastapi_app.py:24: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
/home/satwani/dev/P/DMML_assignment_2/test_by_download/recommendation-pipeline-revising_deduplication_code/recommendation-pipeline-revising_deduplication_code/.venv/lib/python3.14/site-packages/mlflow/server/fastapi_app.py:24: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
2026/11/09 20:18:40 INFO:     Started server process [50031]
2026/11/09 20:18:40 INFO:     Waiting for application startup.
2026/11/09 20:18:40 INFO:     Started server process [50033]
2026/11/09 20:18:40 INFO:     Waiting for application startup.
2026/11/09 20:18:40 INFO:     Started server process [50034]
2026/11/09 20:18:40 INFO:     Waiting for application startup.
2026/11/09 20:18:40 INFO:     Started server process [50032]
2026/11/09 20:18:40 INFO:     Waiting for application startup.
2026/11/09 20:18:41 INFO:     Application startup complete.
2026/11/09 20:18:41 INFO:     Application startup complete.
2026/11/09 20:18:41 INFO:     Application startup complete.
2026/11/09 20:18:42 INFO:     Application startup complete.
2026/11/09 20:18:44 INFO:     127.0.0.1:40114 - "GET / HTTP/1.1" 200 OK
2026/11/09 20:18:45 INFO:     127.0.0.1:40114 - "GET /static-files/static/js/main.00d66947.js HTTP/1.1" 200 OK
2026/11/09 20:18:46 INFO:     127.0.0.1:40120 - "GET /static-files/static/css/main.b0f58ef9.css HTTP/1.1" 200 OK
2026/11/09 20:18:47 INFO:     127.0.0.1:40120 - "GET /static-files/TelemetryLogger.telemetry-worker.d0f8ea2c83c08b63528f.worker.js HTTP/1.1" 200 OK
2026/11/09 20:18:47 INFO:     127.0.0.1:40114 - "GET /static-files/favicon.ico HTTP/1.1" 200 OK
2026/11/09 20:18:48 INFO:     127.0.0.1:40120 - "GET /ajax-api/3.0/mlflow/server-info HTTP/1.1" 200 OK
2026/11/09 20:18:48 INFO:     127.0.0.1:40120 - "GET /static-files/static/js/4205.9e750e7b.chunk.js HTTP/1.1" 200 OK
```

MLflow experiment-tracking setup



MLflow experiment runs

Parameters

Show diff only

algorithm	TruncatedSVD	
dataset	Amazon Reviews 2018 - Software (5-core)	Amazon Reviews 2018 - Software (5-core)
feature_as_of	1443951360.0	1443951360.0
n_factors	40	
n_items	722	722
n_users	1719	1719
ranking		item_popularity (interaction count)
rating_factors	20	
rating_prior_strength		5
seed	42	42
split_cutoff	2015-10-04	2015-10-04

Model parameters recorded in MLflow

ranking		item_popularity (interaction count)
rating_factors	20	
rating_prior_strength		5
seed	42	42
split_cutoff	2015-10-04	2015-10-04
train_data_hash	cb70ab2c4a935623	cb70ab2c4a935623
train_rows	9528	9528

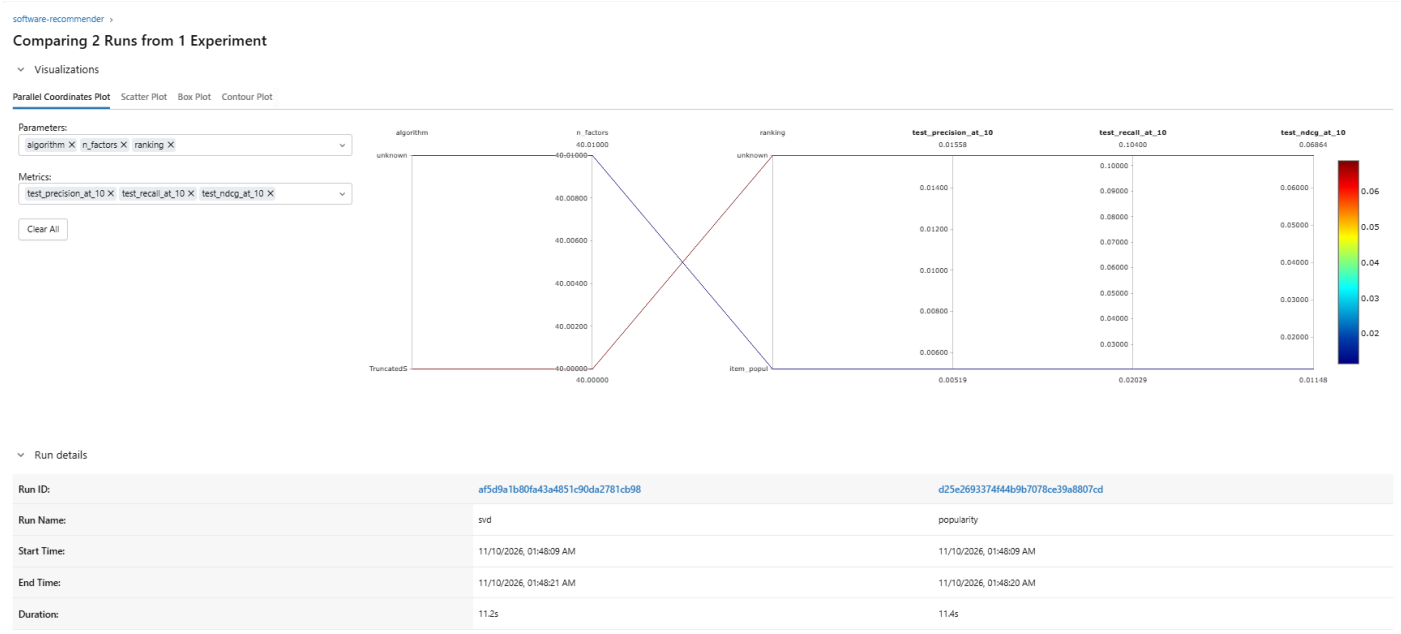
Metrics

Show diff only

fit_seconds	0.698	0.068
test_coverage	0.378	0.021
test_diversity	0.805	0.854
test_eval_rating_pairs	703	703
test_eval_users	308	308
test_latency_ms_per_user	0.029	0.032

Model metrics recorded in MLflow

test_mae	0.964	1.117
test_ndcg_at_10	0.069	0.011
test_precision_at_10	0.016	0.005
test_recall_at_10	0.104	0.02
test_rmse	1.197	1.372



MLflow model-run comparison

```
e$ .venv/bin/python -m src.evaluation.evaluate
2026-11-09 20:18:20,779 | INFO | evaluation | popularity {'precision@10': 0.0052, 'recall@10': 0.0203, 'ndcg@10': 0.0115, 'coverage': 0.0208, 'diversity': 0.8536, 'rmse
': 1.3723, 'mae': 1.1166, 'eval_users': 308.0, 'eval_rating_pairs': 703.0, 'latency_ms_per_user': 0.0324}
2026-11-09 20:18:20,966 | INFO | evaluation | svd        {'precision@10': 0.0156, 'recall@10': 0.104, 'ndcg@10': 0.0686, 'coverage': 0.3781, 'diversity': 0.8047, 'rmse'
: 1.1972, 'mae': 0.9643, 'eval_users': 308.0, 'eval_rating_pairs': 703.0, 'latency_ms_per_user': 0.0288}
2026-11-09 20:18:21,047 | INFO | evaluation | Results written to reports/model_comparison.md
```

Model Evaluation

Model	Precision@10	Recall@10	NDCG@10	Coverage	Diversity	RMSE	MAE
popularity	0.0052	0.0203	0.0115	0.0208	0.8536	1.3723	1.1166
svd	0.0156	0.1040	0.0686	0.3781	0.8047	1.1972	0.9643

Comparison of models' results

7. Orchestration and Scheduling:

```
(venv) satwani@HPE-5CG2500XP6:~/dev/P/DMML_assignment_2/recommendation-pipeline$ .venv/bin/prefect server start

 ___ ___ ___ ___ ___ ___ _____
| _ \ _ \ __| __| __/ __|_   _|
|  _/   / _|| _|| _| (__  | |
|_| |_|_\___|_| |___\___| |_|

Configure Prefect to communicate with the server with:

    prefect config set PREFECT_API_URL=http://127.0.0.1:4200/api

View the API reference documentation at http://127.0.0.1:4200/docs

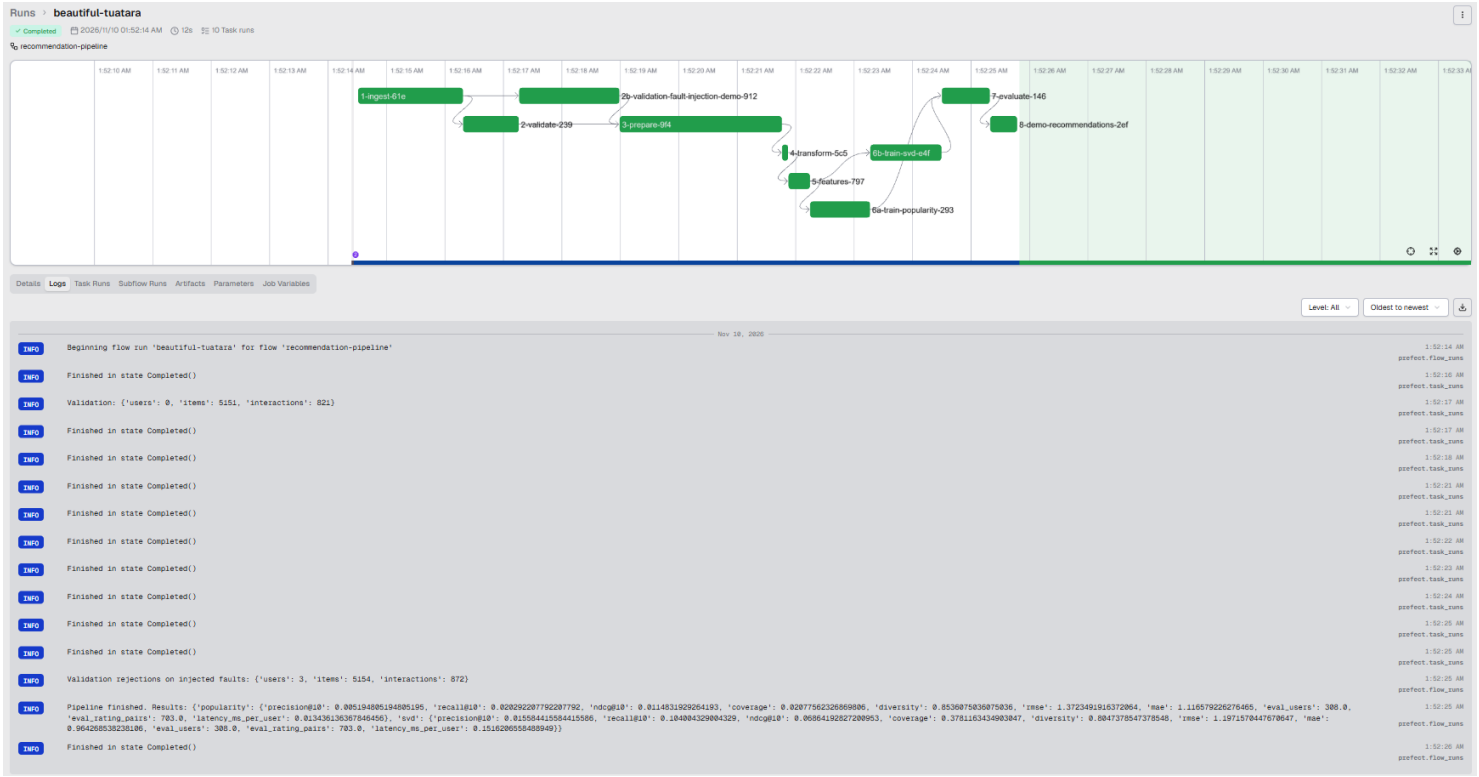
Check out the dashboard at http://127.0.0.1:4200
```

Starting the Prefect Server

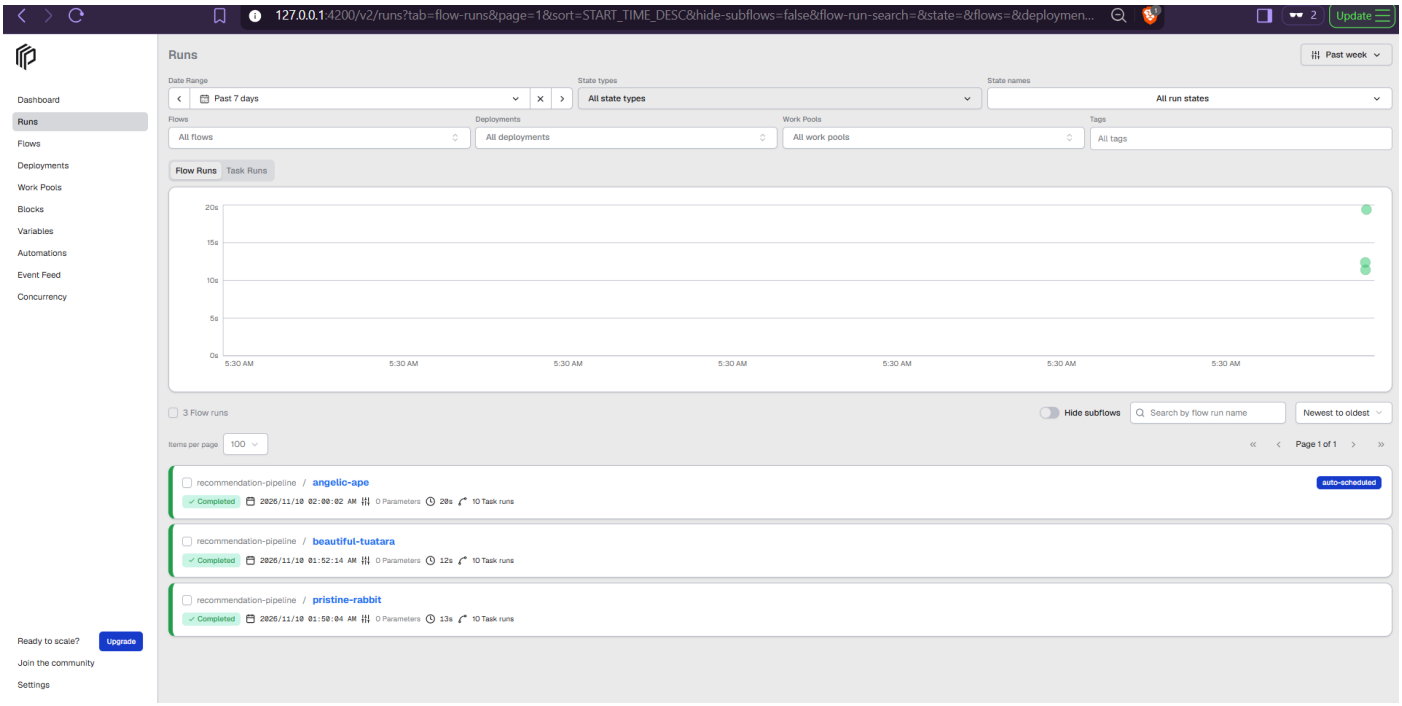
```
.0.0.1:4200/api .venv/bin/python dags/recsys_flow.py
20:22:14.493 | INFO    | Flow run 'beautiful-tuatara' - Beginning flow run 'beautiful-tuatara' for flow 'recommendation-pipeline'
20:22:14.502 | INFO    | Flow run 'beautiful-tuatara' - View at http://127.0.0.1:4200/runs/flow-run/e1180f5b-b2ba-4afe-bb0d-668965796e49
2026-11-09 20:22:14,520 | INFO | ingestion | Using cached download Software_5.json.gz
2026-11-09 20:22:14,521 | INFO | ingestion | Using cached download meta_Software.json.gz
2026-11-09 20:22:15,545 | INFO | ingestion | Wrote users.csv        rows=1826 cols=2
2026-11-09 20:22:16,258 | INFO | ingestion | Wrote items.csv        rows=26790 cols=7
2026-11-09 20:22:16,291 | INFO | ingestion | Wrote interactions.csv rows=12805 cols=7
2026-11-09 20:22:16,310 | INFO | ingestion | Ingestion complete
20:22:16.313 | INFO    | Task run '1-ingest-61e' - Finished in state Completed()
2026-11-09 20:22:16,687 | INFO | validation | Loaded users        rows=1826
2026-11-09 20:22:16,687 | INFO | validation | Loaded items        rows=26790
2026-11-09 20:22:16,687 | INFO | validation | Loaded interactions rows=12805
2026-11-09 20:22:17,038 | INFO | validation | users        valid=1826 rejected=0 {}
2026-11-09 20:22:17,122 | INFO | validation | items        valid=21639 rejected=5151 {"duplicate_row": 5151}
2026-11-09 20:22:17,127 | INFO | validation | interactions valid=11984 rejected=821 {"duplicate_row": 821}
2026-11-09 20:22:17,127 | WARNING | validation | warning interactions.repeated_user_item_pairs = 163
2026-11-09 20:22:17,127 | WARNING | validation | warning items.missing_title = 6
2026-11-09 20:22:17,127 | WARNING | validation | warning items.missing_brand = 659
2026-11-09 20:22:17,127 | WARNING | validation | warning items.missing_categories = 2063
2026-11-09 20:22:17,127 | WARNING | validation | warning items.missing_price = 20182
2026-11-09 20:22:17,127 | WARNING | validation | warning items.unparseable_price = 1290
2026-11-09 20:22:17,127 | WARNING | validation | warning items.missing_description = 3597
2026-11-09 20:22:17,128 | WARNING | validation | warning users.missing_user_name = 1
2026-11-09 20:22:17,265 | INFO | validation | Validation report written to reports/validation_report.md
20:22:17.266 | INFO    | Task run '2-validate-239' - Validation: {'users': 0, 'items': 5151, 'interactions': 821}
20:22:17.269 | INFO    | Task run '2-validate-239' - Finished in state Completed()
2026-11-09 20:22:18,551 | INFO | validation | Loaded users        rows=1829
2026-11-09 20:22:18,551 | INFO | validation | Loaded items        rows=26790
2026-11-09 20:22:18,552 | INFO | validation | Loaded interactions rows=12813
2026-11-09 20:22:18,891 | INFO | validation | users        valid=1826 rejected=3 {"duplicate_row": 3}
2026-11-09 20:22:18,980 | INFO | validation | items        valid=21636 rejected=5154 {"duplicate_row": 5151, "item_id:missing_value": 3}
2026-11-09 20:22:18,986 | INFO | validation | interactions valid=11941 rejected=872 {"duplicate_row": 823, "rating:rating_in_range": 10, "timestamp:timestamp_in_range": 10, "referential_integrity:i
tem_id_not_in_items": 6, "referential_integrity:user_id_not_in_users": 6, "rating:missing_value": 5, "rating:dtype_numeric": 5, "user_id:missing_value": 4, "timestamp:dtype_integer": 3}
2026-11-09 20:22:18,987 | WARNING | validation | warning interactions.repeated_user_item_pairs = 163
2026-11-09 20:22:18,987 | WARNING | validation | warning items.missing_title = 6
2026-11-09 20:22:18,987 | WARNING | validation | warning items.missing_brand = 659
2026-11-09 20:22:18,987 | WARNING | validation | warning items.missing_categories = 2063
2026-11-09 20:22:18,987 | WARNING | validation | warning items.missing_price = 20182
2026-11-09 20:22:18,987 | WARNING | validation | warning items.unparseable_price = 1290
2026-11-09 20:22:18,987 | WARNING | validation | warning items.missing_description = 3597
2026-11-09 20:22:18,987 | WARNING | validation | warning users.missing_user_name = 1
2026-11-09 20:22:18,988 | INFO | validation | Validation report written to reports/validation_report_fault_injection.md
20:22:18.990 | INFO    | Task run '2b-validation-fault-injection-demo-912' - Finished in state Completed()
2026-11-09 20:22:21,591 | INFO | preparation | Duplicates resolved within time partitions: 97 repeated user-item rows removed
2026-11-09 20:22:21,768 | INFO | preparation | Split at 2015-10-04: train=9528 test=2359 (cold users in test: 589 rows)
20:22:21.769 | INFO    | Task run '3-prepare-9f4' - Finished in state Completed()
```

End-to-end orchestration flow

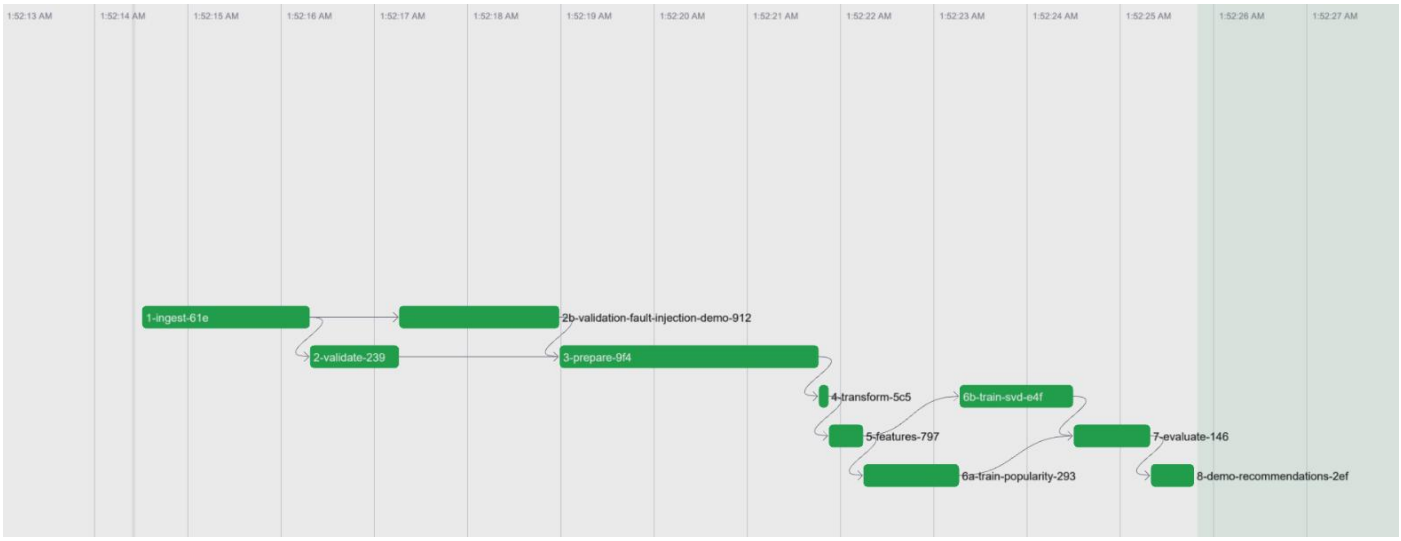
```
20:22:18.990 | INFO    | Task run '2b-validation-fault-injection-demo-912' - Finished in state Completed()
2026-11-09 20:22:21,591 | INFO | preparation | Duplicates resolved within time partitions: 97 repeated user-item rows removed
2026-11-09 20:22:21,768 | INFO | preparation | Split at 2015-10-04: train=9528 test=2359 (cold users in test: 589 rows)
20:22:21.769 | INFO    | Task run '3-prepare-9f4' - Finished in state Completed()
2026-11-09 20:22:21,875 | INFO | transformation | user_activity_summary=1719 rows | item_popularity=722 rows | matrix (1719, 722) density=0.0077
20:22:21.877 | INFO    | Task run '4-transform-5c5' - Finished in state Completed()
2026-11-09 20:22:22,245 | INFO | features | Materialized 22 features: users=1719 items=21639 interactions=9528
20:22:22.248 | INFO    | Task run '5-features-797' - Finished in state Completed()
2026-11-09 20:22:23,277 | INFO | training | Trained popularity in 0.03s -> popularity.joblib (run 323a9f77)
20:22:23.279 | INFO    | Task run '6a-train-popularity-293' - Finished in state Completed()
2026-11-09 20:22:24,499 | INFO | training | Trained svd        in 0.92s -> svd.joblib (run b43cdd48)
20:22:24.501 | INFO    | Task run '6b-train-svd-e4f' - Finished in state Completed()
2026-11-09 20:22:24,934 | INFO | evaluation | popularity {'precision@10': 0.0052, 'recall@10': 0.0203, 'ndcg@10': 0.0115, 'coverage': 0.0208, 'diversity': 0.8536, 'rmse': 1.3723, 'mae': 1.1166, 'ev
al_users': 308.0, 'eval_rating_pairs': 703.0, 'latency_ms_per_user': 0.0134}
2026-11-09 20:22:25,214 | INFO | evaluation | svd        {'precision@10': 0.0156, 'recall@10': 0.104, 'ndcg@10': 0.0686, 'coverage': 0.3781, 'diversity': 0.8047, 'rmse': 1.1972, 'mae': 0.9643, 'eva
l_users': 308.0, 'eval_rating_pairs': 703.0, 'latency_ms_per_user': 0.1516}
2026-11-09 20:22:25,325 | INFO | evaluation | Results written to reports/model_comparison.md
20:22:25.328 | INFO    | Task run '7-evaluate-146' - Finished in state Completed()
2026-11-09 20:22:25,786 | WARNING | inference | Unknown user without usable history -> popularity fallback
2026-11-09 20:22:25,792 | INFO | demo | Wrote sample_recommendations.md
20:22:25.808 | INFO    | Task run '8-demo-recommendations-2ef' - Finished in state Completed()
20:22:25.812 | INFO    | Flow run 'beautiful-tuatara' - Validation rejections on injected faults: {'users': 3, 'items': 5154, 'interactions': 872}
20:22:25.813 | INFO    | Flow run 'beautiful-tuatara' - Pipeline finished. Results: {'popularity': {'precision@10': 0.005194805194805195, 'recall@10': 0.020292207792207792, 'ndcg@10': 0.01148319292
64193, 'coverage': 0.02077562326869806, 'diversity': 0.8536075036075036, 'rmse': 1.3723491916372064, 'mae': 1.116579226276465, 'eval_users': 308.0, 'eval_rating_pairs': 703.0, 'latency_ms_per_user'
: 0.013436136367846456}, 'svd': {'precision@10': 0.015584415584415586, 'recall@10': 0.104004329004329, 'ndcg@10': 0.06864192827200953, 'coverage': 0.3781163434903047, 'diversity': 0.804737854737854
8, 'rmse': 1.1971570447670647, 'mae': 0.964268538238106, 'eval_users': 308.0, 'eval_rating_pairs': 703.0, 'latency_ms_per_user': 0.1516206558488949}}
20:22:26.651 | INFO    | Flow run 'beautiful-tuatara' - Finished in state Completed()
```



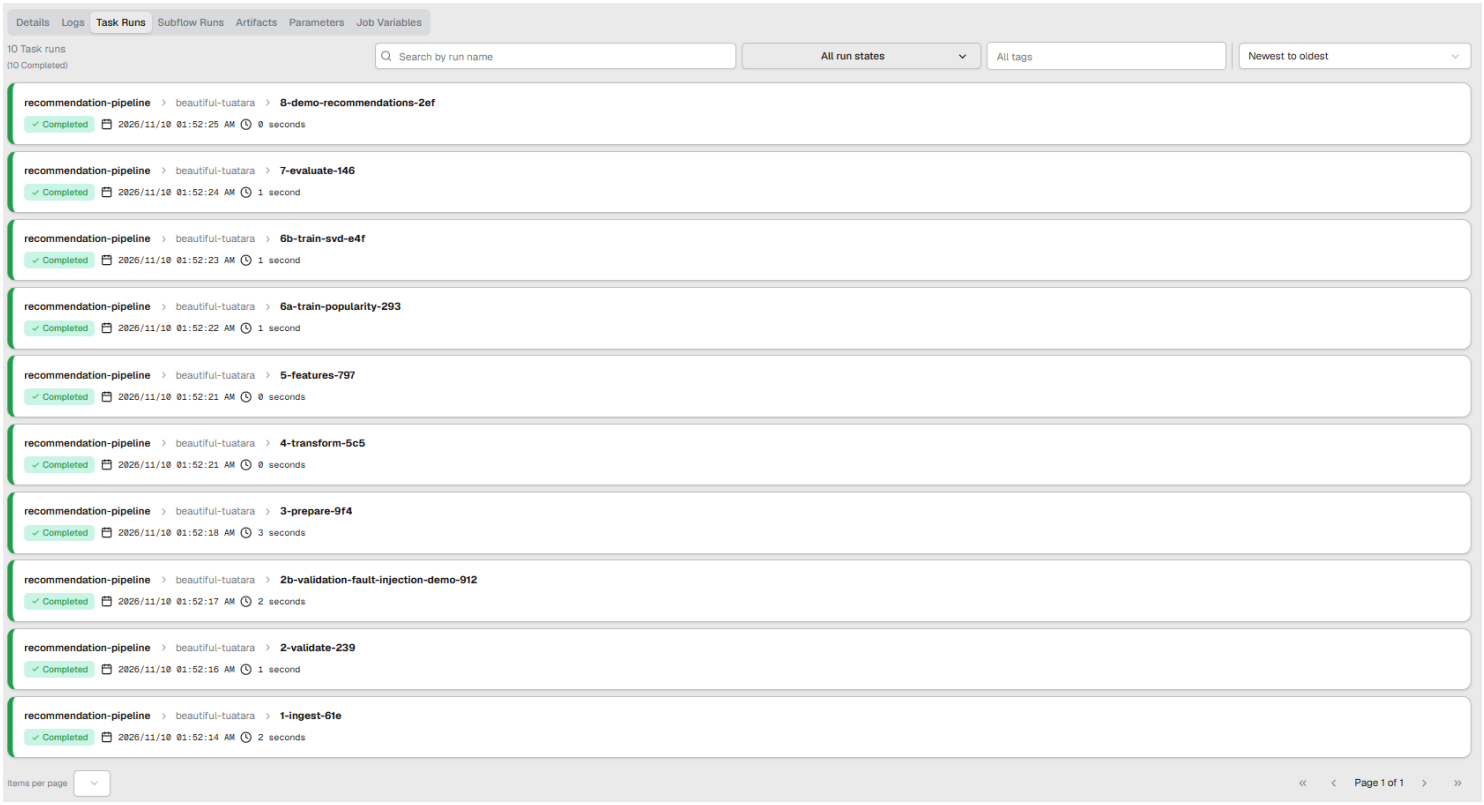
Orchestration successfully completed



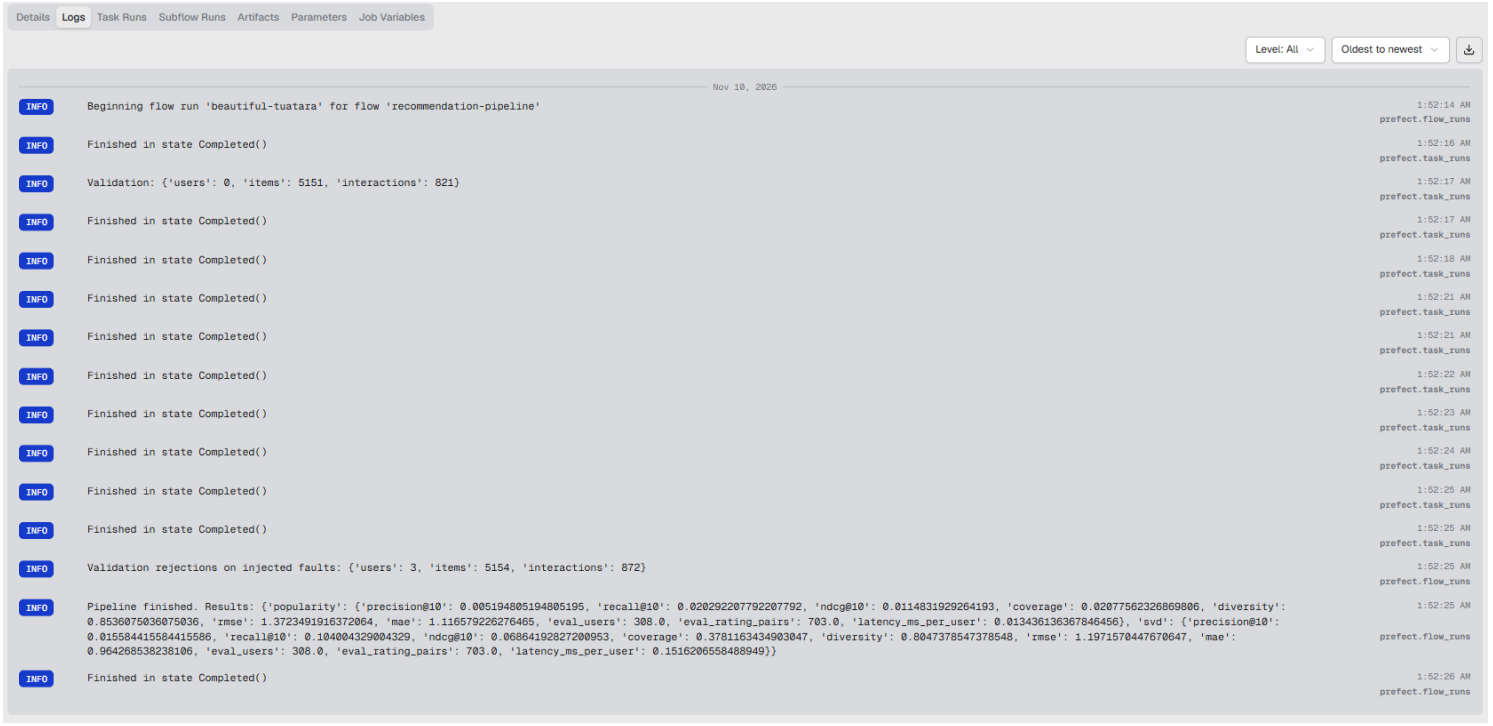
Prefect UI



Pipeline DAG



Prefect task-run details



Prefect flow-run logs

```
e$ PREFECT_API_URL=http://127.0.0.1:4200/api .venv/bin/python dags/recsys_flow.py --serve "30 20 * * *"
Your flow 'recommendation-pipeline' is being served and polling for scheduled runs!

To trigger a run for this flow, use the following command:

        $ prefect deployment run 'recommendation-pipeline/recsys-scheduled'

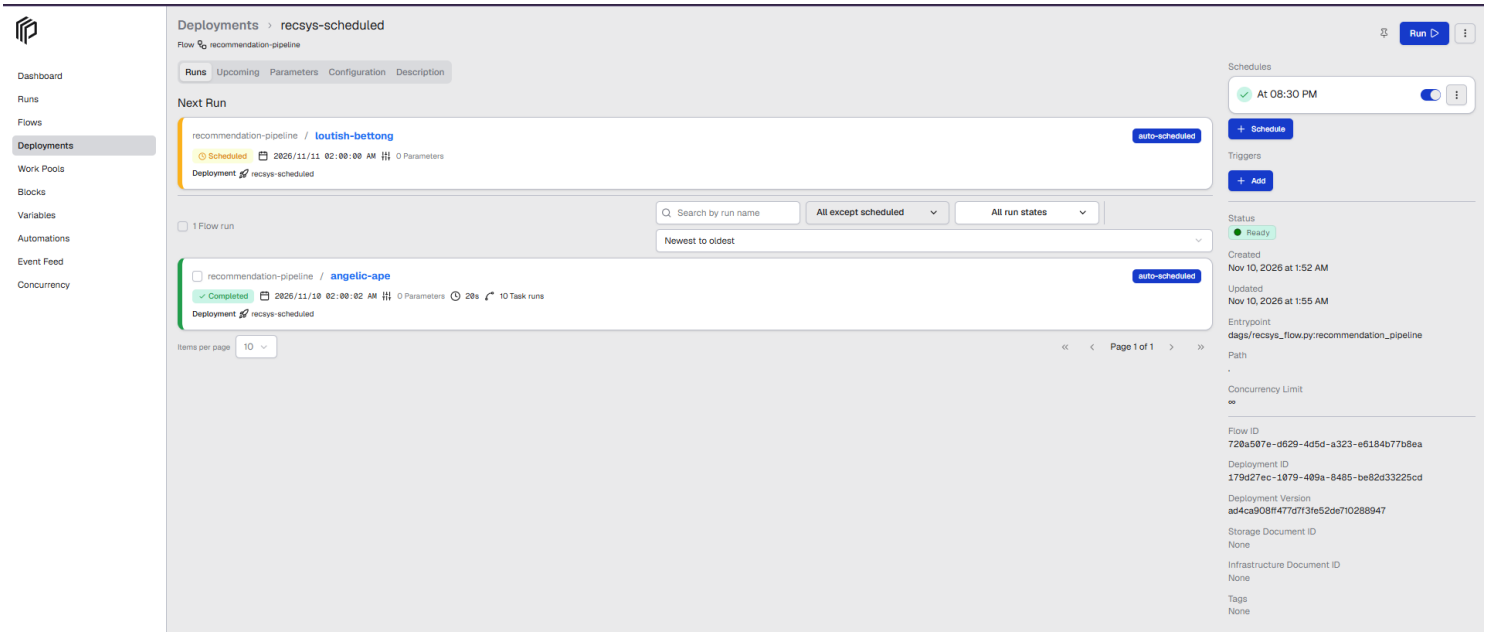
You can also run your flow via the Prefect UI: http://127.0.0.1:4200/deployments/deployment/179d27ec-1079-409a-8485-be82d33225cd

20:30:02.534 | INFO    | Flow run 'angelic-ape' - Beginning flow run 'angelic-ape' for flow 'recommendation-pipeline'
20:30:02.553 | INFO    | Flow run 'angelic-ape' - View at http://127.0.0.1:4200/runs/flow-run/01a1c1fc-6f39-7f3e-b547-95c673bf56d6
2026-11-09 20:30:02,637 | INFO | ingestion | Using cached download Software_5.json.gz
2026-11-09 20:30:02,639 | INFO | ingestion | Using cached download meta_Software.json.gz
2026-11-09 20:30:06,738 | INFO | ingestion | Wrote users.csv       rows=1826 cols=2
2026-11-09 20:30:08,153 | INFO | ingestion | Wrote items.csv       rows=26790 cols=7
```

Cron-based scheduling of the Prefect flow for 20:30 UTC (02:00 IST)



Prefect deployments view



Prefect deployment details showing the runs scheduled for 2:00 AM IST(20:30 UTC)

8. Live Recommendations:

```
ion_code$ .venv/bin/python predict.py --user_id A2THQCHKR89E16 --k 10 --show-history

Past reviews of A2THQCHKR89E16:
             event_time    item_id                                                                                    title     brand            category  rating
2012-12-26 00:00:00+00:00 B0043EV5DU                                                  Norton Antivirus 2011 - 1 User/3 Pc [Old Version]  Symantec Antivirus & Security     3.0
2012-12-26 00:00:00+00:00 B005FIWT74                                                                    Quicken Deluxe 2012    Intuit       Uncategorized     5.0
2012-12-26 00:00:00+00:00 B005S4Y13K TurboTax Deluxe Federal + E-file + State 2011 for PC [Download] [Old Version]    Intuit     Digital Software     5.0
2012-12-26 00:00:00+00:00 B006VRWPYK                                                  Norton 360 6.0 - 1 User / 3 PC [Old Version]  Symantec Antivirus & Security     1.0
2012-12-26 00:00:00+00:00 B01F7RJHIQ       Microsoft Office Home and Student 2010 Family Pack, 3PC (Disc Version) Microsoft    Business & Office     5.0
2012-12-31 00:00:00+00:00 B008RA5A00                                                                    Quicken Deluxe 2013    Intuit       Uncategorized     5.0
2013-03-16 00:00:00+00:00 B009HBCU9W           TurboTax Deluxe Federal + E-File + State 2012 for PC [Old Version]    Intuit     Digital Software     5.0

Top-10 recommendations - source: svd (known user)

 rank    item_id                                                                                    title     brand            category  price  score
    1 B00FFINOWS                                  TurboTax Deluxe Fed + Efile + State 2013 OLD VERSION    Intuit     Digital Software 202.08 1.9070
    2 B00E6LJ2SA                                                Quicken Deluxe 2014 [Old Version]    Intuit       Uncategorized  17.10 1.4913
    3 B00NG7JVSQ TurboTax Deluxe 2014 Fed + State + Fed Efile Tax Software - Win [Downl    Intuit     Digital Software  31.99 0.8160
    4 0763855553 Microsoft Office 365 Home | 1-year subscription, 5 users, PC/Mac Key C Microsoft    Business & Office  90.05 0.6581
    5 B000W3RSGE                                  TurboTax Deluxe Federal + State 2007 [OLD VERSION]    Intuit       Uncategorized  17.10 0.6110
    6 B003YJ5DKG                                              Quicken Deluxe 2011 - [Old Version]    Intuit  Accounting & Finance  45.95 0.6051
    7 B00440D8OW TurboTax Deluxe Federal + e-File + State 2010 [Download] [OLD VERSION]    Intuit     Digital Software  31.99 0.5365
    8 B005CELLHK                                                Quicken WillMaker Plus 2012      NOLO  Accounting & Finance  45.95 0.5336
    9 B0013A5GGE                                  Norton 360 Version 2.0 [Old Version]  Symantec  Antivirus & Security  24.60 0.5118
   10 B004XO6V1G                                                          CorelCAD (Old Version)     Corel Design & Illustration  24.99 0.4608
```

Recommendations generated using the SVD model

```
ion_code$ .venv/bin/python predict.py --user_id A2THQCHKR89E16 --k 10 --show-history --model popularity

Past reviews of A2THQCHKR89E16:
             event_time    item_id                                                                                    title     brand            category  rating
2012-12-26 00:00:00+00:00 B0043EV5DU                                                  Norton Antivirus 2011 - 1 User/3 Pc [Old Version]  Symantec Antivirus & Security     3.0
2012-12-26 00:00:00+00:00 B005FIWT74                                                                    Quicken Deluxe 2012    Intuit       Uncategorized     5.0
2012-12-26 00:00:00+00:00 B005S4Y13K TurboTax Deluxe Federal + E-file + State 2011 for PC [Download] [Old Version]    Intuit     Digital Software     5.0
2012-12-26 00:00:00+00:00 B006VRWPYK                                                  Norton 360 6.0 - 1 User / 3 PC [Old Version]  Symantec Antivirus & Security     1.0
2012-12-26 00:00:00+00:00 B01F7RJHIQ       Microsoft Office Home and Student 2010 Family Pack, 3PC (Disc Version) Microsoft    Business & Office     5.0
2012-12-31 00:00:00+00:00 B008RA5A00                                                                    Quicken Deluxe 2013    Intuit       Uncategorized     5.0
2013-03-16 00:00:00+00:00 B009HBCU9W           TurboTax Deluxe Federal + E-File + State 2012 for PC [Old Version]    Intuit     Digital Software     5.0

Top-10 recommendations - source: popularity (known user)

 rank    item_id                                                                                    title     brand            category  price  score
    1 B0001FS9NE                                  Sony VAIO 802.11a/b/g Ethernet Converter (PCWADE80)      Sony Networking & Servers  26.52  189.0
    2 B000050ZRE                                            Key Span 4-Port USB Hub - White (UH-4WH)   Keyspan Networking & Servers  26.52  189.0
    3 B000EORV8Q                                            Irocks Red USB 2.0 Illuminated 4PORT Hub   Buslink Networking & Servers  26.52  189.0
    4 B0000AZJY6                                  Iogear POWERLINE/HOMEPLUG WALL MOUNT (GHPU21)    IOGEAR Networking & Servers  26.52  186.0
    5 B00EZPXYP4                                                NORTON 360 21.0 EN 1U 3LIC MM  Symantec Antivirus & Security  24.60   96.0
    6 B00F8K9MZQ                                  TurboTax Deluxe Fed + Efile + State 2013 OLD VERSION    Intuit     Digital Software 202.08   70.0
    7 B008H3SW4I                                      Microsoft Windows 8 Pro - Upgrade [Old Version] Microsoft    Operating Systems  11.99   68.0
    8 B00MYXTCGY TurboTax Deluxe 2014 Fed + State + Fed Efile Tax Software [Old Version    Intuit       Uncategorized  17.10   58.0
    9 B00EZQYC8G      Norton Internet Security 2014 - 1 Year / 3 Licenses [Old Version]  Symantec Antivirus & Security  24.60   56.0
   10 B00NG7JVSQ TurboTax Deluxe 2014 Fed + State + Fed Efile Tax Software - Win [Downl    Intuit     Digital Software  31.99   50.0
```

Recommendations generated using the popularity model

```
ion_code$ .venv/bin/python predict.py --history "B00004TYCR:5,B00003JAU7:4" --k 5

Top-5 recommendations - source: svd (fold-in from 2 supplied reviews)

 rank    item_id                                                 title     brand             category  price  score
    1 B00005MOTE Microsoft Windows XP Home Edition Upgrade - Old Version Microsoft     Operating Systems  11.99 0.1330
    2 B00005MOTG Microsoft Windows XP Professional Upgrade [OLD VERSION] Microsoft     Operating Systems  11.99 0.1324
    3 B00E6LJ2SA                        Quicken Deluxe 2014 [Old Version]    Intuit         Uncategorized  17.10 0.1303
    4 B00005MOTH        Microsoft Windows XP Professional [Old Version] Microsoft     Operating Systems  11.99 0.1236
    5 B001TYYZCA                          Corel Painter 11 [OLD VERSION]     Corel Design & Illustration  24.99 0.1208
```

Recommendation example for a new user